בס"ד

החוג לכריית מידע
DEPARTMENT OF DATA MINING
MSc

טבת ה'תשפ"ג

שלב 3

הצעת מחקר בכריית מידע

החוג לתואר שני בכריית מידע

הצעת מחקר לתואר שני

Reference Extraction from Hebrew/Aramaic Reponsa

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המרכז האקדמי לב

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1. Introduction

References are commonly used in rabbinical responsa. This proposed research will attempt to find ways to analyze these references in a way that will allow automatically find the quoted texts.

References in Rabbinical responsa have a few unique characteristics relevant to this proposed research (Hacohen- Kerner et al., 2011), e.g.:

1. While in academic literature the references are written in one of a few structured formats, in rabbinical literature the references are written in a freer text format (many structures are used).
2. While in academic literature, references typically include enough information to uniquely identify the cited work, in Rabbinical responsa, references are often missing information and therefore many of them are ambiguous.
3. Even when a reference includes all the necessary information, it still may not uniquely identify the cited work.
4. Natural Language Processing (NLP) in Hebrew and Aramaic has been studied relatively little compared to English.
5. The morphology of Hebrew/Aramaic is more complex than English's morphology.

Studies by HaCohen-Kerner et al. (2011) and Mughaz et al. (2019) dealt with automatically identifying references in rabbinical responsa texts, our research will attempt to extract the information from these references in a way that will allow us to find the quoted text at its source.

2. Background

Information extraction can be defined as the task of automatically extracting specific fragments of text to defined categories. Various approaches are taken to extract information from texts in English, significant limits apply when analyzing text in non-English languages, specifically in Hebrew or Aramaic.

Information extraction was traditionally done by pattern learning approaches, such as the study by Soderland (1999) and Yangarber (2000) extracting several pieces of information out of news articles or ads (for example, people moving in or out of positions in government and corporations, house rental prices, etc.), transforming the text into a structured format, equivalent to filling slots in a relational database.

Later research by Chieu and Ng (2002) suggests that using a classification approach for information extraction will allow systems to adapt to a new domain simply by using a standard set of features. Han et al. (2013) present a calcification approach extracting information such as title, author, author affiliation, address, email, phone, date, and more from research paper headers using multi-class SVM (Support Vector Machines) using the following rule-based method with the following main ideas:

- Using a collection of 8,441 first names and 19,613 last names, USA state and city names, country names, month names and abbreviations.
- Classifying each line of the header into one or more of 15 classes (title, author, email, phone, keywords, date etc.) An iterative convergence procedure is then used to improve the line classification by using the predicted class labels of its neighbor lines in the previous round.
- Further metadata extraction is done by seeking the best chunk boundaries of each line. The authors found that the use of the structural patterns of the data and domain-based word clustering can improve the metadata extraction performance.

This achieved an accuracy of between 98% and 100% depending on the component.

Non-English and Non-STEM (Science, Technology, Engineering, and Math) publications have additional challenges over English STEM publications, several reasons can be identified:

- Lower proportion of born-digital or digitized publications (Colavizza & Romanello, 2019).
- Variety of morphology, syntax, and semantics of citations, including references usually made in footnotes, not just reference lists (Colavizza & Romanello, 2019).

Boukhers et al. (2021) research information extraction from German publications viewed the pdf as an image and used a computer vision approach to extract details such as title, authors, journal, abstract, date, DOI, address, affiliation, and email addresses. This required creating a training data set, this was done by manually annotating the first page of 100 German publications and automatically generating a corpus of 44k additional images based on the templates derived from the manual annotation. This approach achieved a precision of ~90.7%.

Previous research by HaCohen-Kerner et al. (2011) on Hebrew/Aramaic texts focuses on the identification of sentences containing references, the main stages of their model are as follows:

- Building a list of 1,004 frequent specific citations for supervised learning. This list contains names of authors, their books/responsa represented by various forms, e.g., different spellings, full names, short names, surnames, family names and nicknames with/without titles, abbreviations, and acronyms.
- Cleaning the documents, deleting various types of additions in brackets and redundant spaces that are not part of the original text.
- Calculating features for each of four future sets, orthographic, quantitate, stop word based and n-grams based.

- Combining all feature sets and applying supervised ML methods using only features that InfoGain estimates their weights as non-zero.

The supervised learning is done by comparing words "suspected" to be citations to known citations collected in stage 1 or grammatical extensions of them using suitable prefixes from a list of 38 common rabbinic prefixes. This list is composed of the most common prefixes of known citations.

Later research by Mughaz et al. (2019) focused also on Hebrew/Aramaic documents finding and tagging citations by extracting sentences containing a name of at least one of three famous authors (manually provided). Within these sentences find common patterns of addressing those three authors and with these patterns finding references to various other authors as well.

Abbreviations (shortened or contracted form of a word or phrase, or specifically in Hebrew a representation of a number “Gematria”) and acronyms (a specific type of abbreviation formed from the first letters of a multi-word term) are used in most responsa references.

While acronyms are a relatively recent addition to the English language, first significantly appearing in the 20th century (Mair, 2006). Hebrew has a long history of acronyms, dating back to at least the first century CE (Ravid, 1990). Finding the full phrase when given an abbreviation in Israeli military text in Hebrew is researched by Ravid (1990). Jacobs et al. (2020) investigated acronyms in modern Hebrew building a dictionary reaching an accuracy of 85.03%.

More relevant to this proposed research Hebrew/Aramaic abbreviations were researched by HaCohen-Kerner et al. (2008A), the solution developed was based the “one sense per discourse” which suggests that an author will be consistent and use any given abbreviation with the same meaning each time. 18 different features were defined. Each feature is described as a rule and implemented as a baseline method; each baseline method had one of six different variations. All variant combinations were tested. The best results were achieved by LIBSVM, using a combination of only one variant from each baseline method. This is attributed to preserving the linearity of the different baseline methods. They archived an accuracy of 96.39%. In 2010 HaCohen-Kerner et al. expanded this research, and instead of using “one sense per discourse” and using the baseline methods in their basic form, this reached an accuracy of 98.07%.

A project called “Dicta” (https://dicta.org.il/) deals with vocalization, punctuation, and abbreviations in Hebrew/Aramaic rabbinical texts. Koppel et al. (2020) treat ambiguous abbreviations as sequence prediction problems that relate not to individual words, but rather to sequences. “Dicta” provides online a set of analytical tools for Hebrew text, including adding vocalization (nikud), abbreviation expansion, Bible and Talmud search, transition from printed scanned text to digital text, partitioning any selected text into distinct stylistic components, identifying exact or approximate quotations of biblical and Talmudic sources. Abbreviation expansion is relevant to this study, however in many cases “Dicta” provides more than one expansion which will limit the use of the Dicta tool for this study.

Tokenization is a standard step in any NLP pipeline, Subword tokenizers are the de-facto standard in modern NLP (Devlin et al., 2019; Raffel et al., 2020; Brown et al., 2020). These algorithms only split words into smaller units and cannot handle more complex language phenomena, such as agglutination and consonant mutation. This may be sufficient for languages with simple grammar, like English, but it is not sufficient for languages with more complex grammar. (Clark et al., 2022). Hebrew has complex morphology, and specific grammar rules and domain-specific dictionaries can provide a language/domain-specific tokenization (Hassler et al., 2006).

3. Study Goals

This Study will attempt to extract information from references in Hebrew and Aramaic texts in rabbinical responsa from the beginning of the 19^{th} century and up.

Given a reference in a Hebrew/Aramaic rabbinical responsa text, this study attempts to find an automatic method enable finding the referenced text by breaking down each reference to its components. Table 1 introduces some examples of references and their components, in cases where more than one possible solution is found, a list of possible solutions will be provided ordered by the confidence level of its correctness. Additional examples are given in Appendix A.

Table 1. Examples of references and their components.

References	Expected outcome			
	Component 1	**Component 2**	**Component 3**	**Component 4**
ביבמות דף ס"א ע"א	יבמות	דף סא	עמוד א	
בש"ע יו"ד סוף סימן רמ"א	שולחן ערוך	יורה דעה	סימן רמא	
ובשו"ת פרי הארץ ח"א (חאו"ח סי' ח)	שו"ת פרי הארץ	חלק א	אורח חיים	סימן ח
ברמב"ם פ"ה ממרים הי"א	רמב"ם	ממרים	פרק ה	הלכה יא

An outcome as suggested will enable a few key capabilities:

- Given an indexed corpus, we can find the quoted text (the construction of indexes for all quoted texts is not in the scope of the proposed research). This can be used as a base for potential studies that will identify or correct wrong references.
- Enable various groups such as children, inexperienced and non-expert learners to see the full quotation in the original texts (especially in cases of ambiguous texts and abbreviations).

- Provide an historical prospective to Jewish law as follows:
  - Significant Changes in the number of references to specific sources can reflect historical events, such as the Holocaust, Jewish return to Israel, and the widespread use of electricity.
  - Highlight the impact of Book burnings and Jewish expulsions on Jewish law caused by removing specific books from the possession of Rabbis.

4. Study Plan

This study will use a corpus of responsa in Hebrew/Aramaic from Jewish Rabbis in the past 120 years. Our corpus contains 13,444 files (each files containing one responsa ruling) from 20 different Responsa books. Four examples of Responsa books are provided in Table 2 Appendix B contains full details about these 20 Responsa books.

Table 2. Details of several Responsa books.

Book Name	File count	Total words	Average words in file	Author details
שו"ת החיד"א	8	18,336	2,292	רבי חיים יוסף דוד אזולאי ישראל, איטליה (1724-1806)
זבחי צדק	86	80,374	934.58	רבי עבדאללה בן אברהם סומך עירק (1813-1889)
יביע אומר	842	3,678,311	4,368.54	רבי עובדיה יוסף ישראל (1920-2013)
אגרות משה	1,833	2,288,379	1,248.43	רבי משה פיינשטיין בלארוס, ארה"ב (1895-1986)

Another dataset that we plan to use in this study is a list of over 50,000 Hebrew religious books provided by the Ozar Hachochma project (https://tablet.otzar.org/), which is a digital library of Hebrew religious books. This dataset includes for each book, its name, author name, and publication year. A few examples of books from “Ozar Hachochma” are provided in Table 2. We will explore different filtering options for this list as this list may contain many false positives, as there are books’ names that are common words in use and not books’ names.

Table 3. Books’ examples provided by “Ozar Hachochma”.

Publishing year (in Hebrew)	Author	Book name
תשס"א	רינהולד, יוסף צבי - פוברסקי, דוד - קרליץ, אברהם ישעיהו	דברי חכמים <ישועת דוד>
תשס"ו	קרליץ, שמריהו יוסף ניסים בן נחום מאיר	חוט שני - 16 כר'
תשמ"ט	קרליץ, שמריהו יוסף ניסים בן נחום מאיר	שעורי הגר"נ קרליץ
תשנ"ה	קרליץ, שלמה שמשון בן מאיר	דברי שלמה

A general description of the planned algorithm is given below:

1. Tokenization
2. Abbreviation expansion
3. Named entity recognition
4. Reference Analysis

There is a widespread variety of text preprocessing methods. Examples of basic types are conversion of uppercase letters into lowercase letters, correction of commonly misspelled words, HTML tag removal, punctuation mark removal, reduction of different sets of emoticon labels to a reduced set of wildcard characters, reduction of replicated characters, replacement of HTTP links to wildcard characters, stop word removal, and word stemming. Examples of more sophisticated preprocessing methods are word lemmatization, translation of common slang words into phrases that express the same ideas without using slang, and expansion of abbreviations. (HaCohen-Kerner et al. 2020). In this study, we will apply the preprocessing methods relevant to Hebrew Responsa.

The inputted text will be tokenized into sentences and words, for each sentence, we will go through all words and classify the sentence as one of three categories: reference, suspected reference, or non-reference. Suspected references can include texts that are structured similar to references, but other reference features are missing. For example, the following sentence from the Hida Responsa (שו"ת החיד"א), first chapter:

"ובחפשי באמתחות הפוסקים מצאתי ראיתי להרב מוהר"י די בוטון ז"ל בעדות ביעקב אשר לו סי' ז' הביא תשובת הרא"ש ז"ל".

The last words in the quoted text may be classified as a suspected reference as it includes an author’s name followed by an abbreviation but missing the common reference structure. To classify whether it is a real reference or not, we will use the detection methods developed by HaCohen-Kerner et al. (2011) and Mughaz et al. (2019).

Tokenization methods are usually not equally suited to all languages, the use of any fixed vocabulary may limit a model's ability to adapt (Clark et al., 2022). To improve the tokenization and reference classification in this study, which focuses on a specific domain of text, we can use a domain-specific list of Hebrew/Aramaic book names and authors. Abbreviation expansion can be done with a free tool provided by Dicta, a non-profit organization that uses machine learning and natural language processing tools to analyze Hebrew texts. If this tool is unable to provide a complete expansion, a semi-manual method using predefined tables for common abbreviation will be used as a backup in cases the 'Dicta' tool does not provide a solution or provides too many solutions.

Named entity recognition will focus on names of authors and books. We will use a semi-supervised approach based on an initial list of books and authors manually provided. A bootstrapping technique will be used as follows: The system will search for sentences that contain the given names and try to identify some contextual clues that are common to the examples. The system will then try to find other instances of book and author names that appear in similar contexts. The learning process will add the new book and author names to the list. This process will be re-applied to the newly found names. Using this process, many book and author names of diffrent contexts will be gathered. Experiments with semi-supervised named entity recognition report performances that rival baseline fully supervised approaches (Nadeau and Sekine, 2007).

For each reference or suspected reference, we will generate an output of the reference components as exemplified in Section 3 ("study goals"), in cases where we have a few possible outcomes we will list all options in a descending order according to their relevance.

We will then create a report detailing the outcomes for each tested text that has been analyzed. This report will include the outcomes for each tested text. If multiple solutions are identified a list will be provided ordered by the confidence in correctness.

Success as defined in Section 3 ("study goals") and exemplified in Table 1 and in Appendix A, will be measured as follows. A manual solution will be provided for 200 examples. This will be used as a ground truth. A correct result will be defined as a result matching the manual solution provided. Success will be defined as correct result provided as the first or only option, and partial success will be defined as an output of a correct result in one of the top 5 possible solutions.

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Appendix A: Example of study suggested output

Text:

שו"ת יביע אומר חלק ז - אורח חיים סימן לו ד"ה ג) אמנם

ג) אמנם ראיתי בשו"ת לבושי מרדכי תליתאה (חאו"ח סימן כה) שנשאל בנ"ד, והשיב, הנה מ"ש מעלתו (הרב השואל) שאין בהזזת כפתור החשמל אפילו משום גרם כיבוי, שאינו אלא כמי שמרחיק את הבגד הסמוך לאש כדי שלא תאחז בו האור, בזה נראים דבריו. אבל מ"ש לענין איסור מוקצה שאין זה בכלל טלטול אלא נגיעה בלבד שמותרת, וכמו שהתיר הרמ"א בסי' שח סעיף ג', לפענ"ד נראה שזה נחשב כטלטול, דלא גרע ממ"ש הרמ"א בהגה (סימן רסה ס"ג) שלא ליגע בנר דולק שהוא תלוי, משום שעל ידי כך הוא מנענע אותו, ואם תנועה בלבד חשובה כטלטול ואסורה, כל שכן הזזת כפתור החשמל. ע"כ. ודבריו תמוהים, שהרי לא אסר הרמ"א אלא בנר הדולק שמא יתנדנד הנר "ויטה", והיינו דהוי כמבעיר או כמכבה, אבל אם היה הנר כבוי מותר, ולא חיישינן משום מוקצה. וכמ"ש המשנה ברורה הנ"ל, והוכיח במישור כן מהאור זרוע, מקור דין זה, וא"כ משם ראיה דלא חיישינן במחובר משום איסור מוקצה, וכמ"ש הרב תורת שבת מילתא בטעמא. ומ"ש עוד הרב לבושי מרדכי שם, שיש לדון להחמיר עוד לפי מ"ש בגמ' ר"פ הבונה (קב ב), היכא דעייל שופתא בקופינא דמרא (שהכניס יתד קטן בתוך בית יד של קרדום להדקו שלא יצא. רש"י). חייב משום בונה או משום מכה בפטיש, ובתוס' שם משמע שהוא הדין לענין סתירה, וא"כ גם בנידון זה מאן לימא לן שאין בזה משום שנראה כסותר או כבונה וכו'. ולכן לפע"ד אין היתר בזה. ע"כ. הנה ידוע שגם הגאון החזון איש (חאו"ח סימן נ אות ט) כתב כיו"ב, שבהדלקת החשמל בשבת יש משום תיקון מנא, שמעמידו על תכונתו להזרים אליו זרם החשמל בתמידות, וקרוב הדבר שזהו חשוב כבונה מן התורה, וכעושה כלי, וכ"ש בזה שכל חוטי החשמל מחוברים לבית, והוה ליה כבונה במחובר, שיש בו משום בנין וסתירה, ואפילו בכלים כיוצא בזה חשיב בונה. ע"ש. אולם מדברי כל האחרונים לא משמע כן, וע' בשו"ת מהרש"ם ח"ב (סימן רמו) שכתב, ולא אכחד כי חוכך אני בלבי בדין הבערת וכיבוי החשמל בשבת, אם הוי מלאכה דאורייתא, כיון שלא היה כמוהו במשכן, ועוד שאינו כאש ממש, שאינו שורף חוט הברזל וכו'. ע"ש. גם בשו"ת מי יהודה (חאו"ח סימן לה) העלה שאיסור הדלקת החשמל בשבת הוי רק מדרבנן. ע"ש. והלבושי מרדכי עצמו בח"א (סימן מז - מח) כתב שאין בהדלקת החשמל איסור מן התורה אלא רק מדרבנן. ע"ש. ואף שבשו"ת קרן לדוד (סימן פ) כתב לחלוק עליו בזה, הנה בהסכמתו של הרב לבושי מרדכי לספר קרן לדוד כתב להשיב על דבריו, ועשה סניגורין לדבריו הראשונים. ע"ש. ואפי' למאי דקי"ל כדעת רוב האחרונים שהדלקת החשמל אסורה מן התורה, וכמו שכתבתי בשו"ת יביע אומר (ח"א חאו"ח סי' יט אות יד), בשם הרבה אחרונים חבל נביאים. ע"ש. מכל מקום לא אשתמיט לשום פוסק מן האחרונים לחדש חידוש גדול כזה שיש בו משום בנין וסתירה. וכן שמעתי בימי חרפי ממורנו הרב הגאון רבי עזרא עטייה זצ"ל, שסברת החזון איש בזה אינה מחוורת. והן עתה ראיתי לידידי הגאון רבי שלמה זלמן אוירבך בשו"ת מנחת שלמה (סימן יא עמוד צא) שהאריך למעניתו לפקפק על דברי החזון איש הנ"ל. ושם בהערה א' הביא ג"כ דברי הרב לבושי מרדכי תליתאה הנ"ל, וכתב עליו, ותמיהני טובא שמכיון שהוא עשוי תמיד לפתוח ולסגור, איך אפשר לחושבו כבונה בשעה שאין זרם בחוטי החשמל ולא נעשה שום דבר בהם כלל. והא לא עדיף מכיסוי בור ודות שיש להם בית אחיזה. ועוד שהוא עצמו נשא ונתן בארוכה בח"א (סימן מז) בעיקר הדין של הדלקה וכיבוי בחשמל, והפריז על המדה לומר דהוי רק מדרבנן, ולמה לא כתב שיש בזה עכ"פ משום בונה וסותר, וקל וחומר הוא מנידונו שם, דהכא הרי מעשהו בשעה שאין זרם בחוטי החשמל, ולא נעשה בהם שום דבר כלל. ע"כ. [וכ"כ עוד בקובץ מאמרים (עמוד ס). ושם ג"כ בא לו בארוכה על דברי החזון איש הנ"ל. ע"ש]. הילכך אין לחוש לזה כלל. [וע' בשו"ת שבט הלוי ח"א (חאו"ח סימן קכא) שכ', דמה שחידש החזון איש שבהדלקת וכיבוי החשמל יש משום בונה וסותר, הוא צ"ע. ע"כ. וע"ע בשו"ת תפלה למשה ח"ב (סימן כ עמוד ר' והלאה) שג"כ ישב על מדוכה זו, ובסוף דבריו הוכיח ממה שכתבנו בשו"ת יביע אומר ויחוה דעת בכמה מקומות דלא שמיעא לי כלומר לא סבירא לי סברת החזון איש הנ"ל. והצדק עמו. ואכמ"ל.

Table 5 contains two examples of expected results from the Respona presented above.

Correct/incorrect is based on a manual solution treated as a ground truth. N/A (Not available) is marked when the reference was not entered in the manual solution list.

Table 5. Examples of expected output.

Row in text	Reference	Option	Component 1	Component 2	Component 3	Component 4	Success test
1	בשו"ת לבושי מרדכי תליתאה (חאו"ח סימן כה)	1	שו"ת מרדכי	מהדורה תליתאה	חלק אור החיים	סימן כה	Correct
		2	שו"ת מרדכי תליתאה	חלק אור החיים	סימן כה		Incorrect
32	בשו"ת תפלה למשה ח"ב (סימן כ עמוד ר' והלאה)	1	שו"ת תפלה למשה	חלק ב	סימן כ	עמוד ר	Correct

המרכז האקדמי לב

החוג לכריית מידע
DEPARTMENT OF DATA MINING
MSc

Appendix B: Details about our Current Corpus of Responsa books

	Book name	Author details	Total word count	File count	Average words per file
1	אבני נזר	הרב אברהם בורנשטיין פולין (1838-1910)	1,609,659	1,454	1,107.06
2	אגרות משה	הרב משה פיינשטיין בלארוס, ארה"ב (1895-1986)	2,288,379	1,833	1,248.43
3	דעת כהן	הרב אברהם יצחק הכהן קוק לטביה, ישראל (1865-1935)	263,564	242	1,089.11
4	החידא	הרב חיים יוסף דוד אזולאי ישראל, איטליה (1724-1806)	18,336	8	2,292.00
5	הר צבי	הרב צבי פסח פראנק ליטא,ישראל (1873-1960)	652,306	722	903.47
6	זבחי צדק	הרב עבדאללה בן אברהם סומך עירק (1813-1889)	80,374	86	934.58
7	יביע אומר	הרב עובדיה יוסף ישראל (1920-2013)	3,678,311	842	4,368.54
8	יחוה דעת	הרב עובדיה יוסף ישראל (1920-2013)	766,665	444	1,726.72
9	יחל ישראל	הרב ישראל מאיר לאו ישראל	228,665	91	2,512.80
10	מנחת יצחק	הרב יצחק יעקב ווייס אוקראינה , ישראל (1902-1989)	2,240,956	1,471	1,523.42
11	מנחת שלמה	הרב שלמה זלמן אויערבאך ישראל (1910-1995)	815,487	231	3,530.25
12	משפטי עוזיאל	הרב בן-ציון מאיר חי עוזיאל ישראל (1880-1953)	635,058	294	2,160.06
13	נושא האפוד	הרב דוד פיפאנו ייוון, בולגריה (1866-1924)	128,730	34	3,786.18
14	עטרת פז	הרב פנחס זביחי ישראל	1,297,273	319	4,066.69

15	ציץ אליעזר	הרב אליעזר *יהודה וולדנברג* ישראל *(1915-2006)*	1,805,450	815	2,215.28
16	קול מבשר	הרב משולם ראטה אוקראינה, ישראל (1875-1962)	386,631	135	2,863.93
17	רב פעלים	הרב יוסף חיים מבגדאד עיראק (1833-1909)	819,726	519	1,579.43
18	שבט הלוי	הרב שמואל הלוי וואזנר אוסטריה, ישראל (1913-2015)	1,169,889	1,553	753.31
19	שואל ונשאל	הרב כַלְפוֹן משה הכהן תוניסיה (1874-1950)	889,833	1,827	487.05
20	תורה לשמה	הרב יוסף חיים מבגדאד עיראק (1833-1909)	216,614	524	413.39